

Change-log and Notes for revisions to Annotation Scheme v.1.0

Jade Oakes, Anaelle Small, Jiayi Yao

April 1, 2025

Change-log:

Data

- Reduced size of the dataset we will be working with
 - Selected only written text data
 - Selected only Spanish L1 speaker data
 - Extracted only the responses to the essay questions from each data sample and limited our selection based on file size (eliminated longer text selections)
- Created balanced subset of data so proficiency levels are more equally represented
- Decided on data splits for annotators

Labels

- Added label hierarchy
- Changed the order of how the labels are listed in the guidelines
- Added pronoun_case_error
- Separated punctuation_error and capitalization_error
- Removed dependencies
- Changed label to verb_tense_agreement_error
- Change label to subject/object_omission_error

Guidelines

- Added negative examples for each label
- Defined measurements for determining proficiency level based on number and types of errors
- Added an introduction paragraph
- Revised annotation goal section
- Added Label Studio Software section, including links to Label Studio download and documentation and technical setup instructions
- Added File Organization section with links to data splits (Google Drive) and specifications for file naming conventions

Notes for revisions while we annotate:

Anaelle

here are some of my preliminary notes from doing a couple annotations:

- possible confusion with conjugation tags and agreement tags as they occur in the same position. we should give more detail to differentiate them for annotators.
- semantic errors - i think these should only be reflected in document level proficiency rating
- dependencies are a little weird in label studio - it looks like the second token is the dependent. we should give more detail on how to do this to minimize errors. maybe distinct tags for subject and verb?
 - Yes we should def work out how to do the dependencies (if we still want to do them)
- in a structure like "one bottles" - definitely this is an article error, but should we also say its an agreement error? i think no because its not a verb. we should clarify this in the schema
 - Jade - I think having an error label hierarchy might be a way to handle this? The same for the multiple labels discussion below
- how to deal with spelling errors that are so severe that the intended meaning cannot be deduced, therefore making it difficult to annotate for errors while ignoring spelling?
 - Jade - My initial thought is to just skip samples like that. Idk how we would specify that in the guidelines - maybe we would just say something like "For any sample, if the intended meaning cannot be deduced due to the severity of the errors, you may skip that entire sample." Would we still want them to assign a proficiency level or no?
- what to do when errors are difficult to parse because theres so many? this may be a good criterion for beginner level labels, but makes it difficult to annotate
- example: true id 1
- looks like it may not be possible to given tokens multiple labels - could be a problem with pre-annotated errors or maybe if a token is a part of multiple errors
 - Jade - I tried assigning two labels to the same token and it seems to have worked. It blends the colors of the labels together, so visually it might make it difficult to see which labels are which. We could limit one label per token and list the labels as a hierarchy for which to prioritize?
- I don't think we need to have three separate categories for conjugation errors, sv agreement errors, and number agreement errors. sv agreement can probably be put into one of those categories. also may cause ambiguity with articles - what is an article error vs what is a number agreement error

Jade

- "taked" labeled as spelling_error, however should it be a conjugation/tense_error instead? error label hierarchy

- I did not add "colours" to the British spellings list so it is marked as a spelling_error, but I think this is insignificant when we consider how much data we have
- There are some spelling, punctuation, and capitalization errors that did not get pre-annotated. I think generally we should have annotators ignore those cases and focus more on the grammatical errors, but if they see something like "taked" marked as a spelling error they could potentially remove the spelling_error label and assign the "conjugation/tense_error label (error label hierarchy)
- "other" instead of "another"
- Spanish names: <https://github.com/jvalhondo/spanish-names-surnames?tab=readme-ov-file>
- British spellings: <https://gist.github.com/ZekNikZ/5e7dd531df99be4408bd768ded36aad9>
- note: C1 cut off at 1.3 KB = 107 samples
- note: C2 cut off at 2 KB = 27 samples
- note: A1 - B2: randomize each set of files, then select 140 per annotator?
 - A1 = 35
 - A2 = 35
 - B1 = 35
 - B2 = 35
 - C1 = 35
 - C2 = 9
- add word_choice_error (not doing this)
 - He found a trunk and tried to catch it.
 - Anaëlle: this one could be pretty subjective, i'm not sure i agree that saying "a big goodbye" is an error
- add pronoun_case_error
 - and Snow followed he.
 - Anaëlle: I found this example, should we classify cases like these as a pronoun error? same with referring to charlie chaplin with the feminine pronoun?
He was just calling it name
- for conjugation/tense_error, specify tagging all words that are part of the verb
- other?
 - a boy was with his dog and a frog intro a jar.
- move pre-annotated errors to the end of the Error Types section?
- should we change conjugation/tense_error to verb_tense_error? the conjugation part is confusing bc we also have a subject-verb_agreement_error

- maybe consider using percentage of error types present as the metric for measuring proficiency level (vs error type counts bc each text contains a different number of tokens)

Jiayi

- preposition or word choice? Look a place -> look for a place/look the place
- Reference? "He gave him the baby" "he" and "him" refer to two men
- "baby car" -> "baby-carriage"?
- owned to verb tense? preposition?

Notes for revisions from meetings:

- annotating dependencies
- could pre-annotate spelling and punctuation errors
- semantic errors category potentially
- have annotators add the correct version
- have annotators assign a level to the document along with the errors
- think about what we could give an annotator and what we could give them
- inner annotator agreement and calculate a score (or something) and see if people can identify the correct proficiency level with little training
- identify edge cases before handing it off
- should cut dataset down a bit
 - can limit selections to ones that are a certain length
- look into writing Python code to tag the spelling, punctuation, and capitalization errors to match the labels in Label Studio
- add into guidelines to have annotators not worry about annotating spelling, punctuation, and capitalization errors
- pre-tagging - try it and see if it works, just make it clear to annotators what we need annotated and what we don't
- reorder errors in the guidelines for hierarchy
- specify that there is an order of priority by which to assign labels
- what are negative errors for our scheme? we thought we had included enough info; including an example with no errors would just be a correct sentence, and we don't see how that would be especially helpful for our task
- get rid of the dependencies
- "other" label needs more examples
- tweak the descriptions of the labels - more clear and specific to the error
- write the "capitalization_error" description